

TECHNICAL SPECIFICATIONS

“Supply of E-Bicycle testing equipment with required fixtures for meeting electrical safety standards and performance requirements of e-bicycles and their sub-assemblies”

UNIDO Project “Improving quality, productivity and innovation for enhanced competitiveness of the Indian bicycle and bicycle parts industry” (No. 200297)

1. Background Information

The United Nations Industrial Development Organization (UNIDO) is the specialized agency of the United Nations that promotes industrial development for poverty reduction, inclusive globalization and environmental sustainability. The mission of UNIDO, as described in the *Lima Declaration* adopted at the fifteenth session of the UNIDO General Conference in 2013 as well as the *Abu Dhabi Declaration* adopted at the eighteenth session of UNIDO General Conference in 2019, is to promote and accelerate inclusive and sustainable industrial development (ISID) in Member States. The relevance of ISID as an integrated approach to all three pillars of sustainable development is recognized by the 2030 Agenda for Sustainable Development and the related Sustainable Development Goals (SDGs), which will frame United Nations and country efforts towards sustainable development. UNIDO’s mandate is fully recognized in SDG-9, which calls to “Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation”. The relevance of ISID, however, applies in greater or lesser extent to all SDGs. Accordingly, the Organization’s programmatic focus is structured in four strategic priorities: Creating shared prosperity; Advancing economic competitiveness; Safeguarding the environment; and Strengthening knowledge and institutions.

Each of these programmatic fields of activity contains a number of individual programmes, which are implemented in a holistic manner to achieve effective outcomes and impacts through UNIDO’s four enabling functions: (i) technical cooperation; (ii) analytical and research functions and policy advisory services; (iii) normative functions and standards and quality-related activities; and (iv) convening and partnerships for knowledge transfer, networking and industrial cooperation. Such core functions are carried out in Departments/Offices in its Headquarters, Regional Offices and Hubs and Country Offices.

The Directorate of Technical Cooperation and Sustainable Industrial Development (TCS), headed by a Managing Director, ensures the Organization's application of strategies and interventions for sustainable industrial development related to environment, energy, Micro, Small and Medium-Enterprises (MSMEs), and digitalization. The Directorate also oversees the Organization's normative contribution to achieving the Sustainable Development Goals through industrial policy advice and capacity development. Through coordination in-house and with Member States and industry stakeholders, it ensures that the services provided in these areas contribute to effective and appropriate technical, business and policy solutions and are focused on results, scaling up and positioning UNIDO as a leading platform for industrial development in developing countries and global fora. The Directorate is responsible for the Division of Industrial Policy Advice and Capacity Development (TCS/IPC), and technical Divisions of Circular Economy and Green Industry (TCS/CEG), Energy and Climate Action (TCS/ECA), Climate Innovation and Montreal Protocol (TCS/CMP); MSME Competitiveness, Quality and Job Creation (TCS/SME); and Digital Transformation and Artificial Intelligence (TCS/DAI). Leveraging the diverse skill sets of UNIDO personnel and the services provided by the two TC directorates, TCS collaborates closely with IET to develop and implement programmes and projects, aiming at enhancing synergy and complementarity and maximizing UNIDO corporate performance and

impacts on the ground. The Directorate also ensures close coordination and collaboration among the Divisions as well as with relevant entities in all Directorates across the Organization.

The Division of Digital Transformation and Artificial Intelligence (TCS/DAI) promotes digital transformation and Artificial Intelligence as well as associated technologies as part of innovation ecosystems consisting of productive capacities in manufacturing, services and digital firms, with the purpose to advance the competitiveness of industries and manufacturing firms in Member States. It helps industries to benefit from the rapid progress in digital and convergent technologies associated with the Fourth Industrial Revolution (4IR), ensuring a smooth transition towards safe and secured cyber-physical industrial systems and a smart society and mitigating any negative adverse effects on employment and quality of work. Through the Division's services, productive transformation is the key, by integrating industrial businesses, dynamic entrepreneurship, and acceleration services, and by applying technological innovation to smart manufacturing, smart energy, smart agribusiness and others, in developing countries, transition and emerging economies, with emphasis in developing countries to provide the opportunity to technologically catch up instead of falling behind industries in more advanced countries.

One of the projects implemented by UNIDO in India is a project titled “Improving quality, productivity and innovation for enhanced competitiveness of the Indian bicycle and bicycle parts industry” (SAP ID: 200297), supported by the Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry, Government of India.

2. Background information relevant to the requirement

The overall objective of the bicycle project is to contribute to enhancing the growth and competitive position of the Indian bicycle industry, globally and domestically. This is foreseen to be achieved by improving performance of industry units, especially MSMEs, regarding quality, productivity, design and innovation, and their adherence to product and performance standards for bicycles and bicycle parts in India. The project adopts a two-pronged, market-oriented approach, as described below:

- Firstly, the project will establish a supply of advanced and diversified quality/safety and e-bike testing facilities and related training and advisory services. At the request of the industry associations, this will be done in the existing technical institution (the Research and Development Centre for Bicycle and Sewing Machine, RDCBSM), with strong industry involvement.
- Secondly, the project will foster demand for these quality and e-bike testing and training and advisory services, by improving firm level manufacturing competencies, in the areas of quality management, productivity, design and innovation. The capacity building activities will focus on enhancing the productivity and innovation capabilities of units through workshops, counselling support and self-learning modalities on key methods and tools.

The two sets of activities of the project are inextricably linked: the creation of testing infrastructure would facilitate the enabling conditions required for the issuance of mandatory safety standards and thereby ensuring product quality and safety of cyclists; the technical capacity building for firms, especially the MSMEs, would equip manufacturers to upgrade production processes in line with requirements set forth by product standards and market requirements, thereby also ensuring the fullest utilization of the established testing infrastructure.

Within the first component of the project, UNIDO aims to build the capabilities of RDCBSM as the nodal technical agency for the industry, to become a vibrant, growth-driven and self-sustaining entity providing testing and certification services as per the evolving needs of the industry.

The testing and calibration facilities of the RDCBSM are duly accredited by the National Accreditation Board for Testing and Calibration Labs (NABL; ISO/IEC 17025) and the Bureau of Indian Standards (BIS). As per the NABL accreditation, the RDCBSM labs carry out mechanical testing of metals and alloys, chemical analysis (Fe, Cu and Al, base metals and alloys), calibration of measuring instruments and gauges and the safety testing of bicycles/bicycle sub-assemblies/parts as per ISO 4210 (parts 1-9), IS 10613 and IS 8098.

RDCBSM requires certain additional equipment and accessories to complement the existing testing equipment for meeting safety standards and performance requirements of the bicycles and its sub-assemblies as well as provide testing support for testing of e-bicycles.

3. Scope of Supply

The equipment and associated services within the scope of this tender aim for testing of electrical safety and performance requirements of e-bicycles and sub-assemblies of e-bicycles under the standard of EN 15194 (and equivalent) for EPAC (electric power assisted cycles) used for city and trekking including folding bicycles. Mountain and racing bikes are excluded.

UNIDO seeks to procure specific equipment to support RDCBSM for carrying out mandatory electrical safety and performance testing of EPACs and their sub-assemblies. The details of the equipment required are indicated in Annex I of this document.

The contractor is expected to provide the following (detailed technical specifications are outlined re in **Annex I**):

- **Supply of equipment** with fixtures, standard spare parts and services as specified in Annex I, which must be properly packaged for transportation and unloading at RDCBSM, Ludhiana (Punjab, India). Application Programming Interface(API) access or data logging interfaces are provided wherever applicable, to enable automation and workflow synchronization.

It should be ensured that the equipment and software included are modular or scalable to support future upgrades or protocol extensions without requiring full system replacement.

- **Transport and delivery** of all items included in the scope of supply at RDCBSM, Ludhiana (Punjab, India);
- **Installation, commissioning, and operationalization** of the provided equipment and fixtures at RDCBSM, Ludhiana. UNIDO requires that operation of the equipment be monitored for a tentative period of 1-2 weeks (subject to change) at the site (RDCBSM), within a period of one (01) month from the delivery of the equipment, fixtures and standard spare parts (as per the scope of supply of these Technical Specifications, to ensure seamless testing);
- **Training** in the operation of the equipment and software after installation. Supplier's personnel may be deployed at the site – RDCBSM, to oversee installation and operation for the requisite period;
- **Documentation:** Operating principles manual as per applicable testing procedures and calibration certificate is to be provided. Test equipment layout is to be included as well;
- A brief report of the demonstration and training activity at the site.

PARTIAL OFFERS FOR COMPLETE LOTS WILL BE ACCEPTED.

As part of the delivery the contractor shall be responsible for the following:

A- Inspection, Check and Testing

- a) The Equipment may be subject to inspection (optionally through online/visual method) by RDCBSM, UNIDO or its authorized representatives and statutory testing agencies if deemed required, at any time prior to shipment and/or dispatch, as well as final inspection within a reasonable time after arrival at the site. The Contractor will also bear the expenses concerning preparation and rendering of inspection-related tests and furnish test reports of trial runs before dispatch of equipment requested by UNIDO;

Such inspection, or its non-performance, shall in no way relieve the Contractor of any responsibility or liability with respect to such equipment nor prejudice against the right of UNIDO to reject unsuitable equipment after arrival at the destination. Unless specifically stated to the contrary in the Purchase Order, all expenses relevant to the preparation and performance of testing, inspection and expediting and the preparation of any test reports or certificates shall be borne by the Contractor.

B - Transportation and delivery

- a) The contractor shall be responsible for the transportation of the equipment/items and their delivery at RDCBSM, Ludhiana (in Punjab, India). All testing equipment requested should be suitably packed for transportation and **shipped within four (04) months from the date of the counter-signed contract**, to be followed by **installation, commissioning, operationalization and training within six (06) months from the signing of the contract.**
- b) Delivery should be in accordance with Delivered at Place (DAP) Incoterms 2020.

Further details on delivery address and consignee will be provided by UNIDO closer to the time of delivery.

C - Insurance

The contractor shall be responsible for providing insurance at 110% of value of the equipment shipped covering the transportation period up to destination.

D - Installation, commissioning, and operationalization

The contractor shall be responsible for installation, commissioning, and operationalization of the provided equipment for testing at RDCBSM. UNIDO requires that the equipment is to be operated for a tentative period of 1-2 weeks (subject to change) at the site (RDCBSM), within a period of one (01) month from the delivery of the equipment, fixtures and standard spare parts (as per the scope of supply of these technical specifications) at RDCBSM, to ensure seamless testing.

E - Training

After the installation and commissioning of equipment, the contractor is expected to provide on-site training for RDCBSM personnel for carrying out trial testing as per EN:15194 standards for EPACs and their sub-assemblies.

F - Warranty

The contractor should provide a **minimum of 2-year (24 months) warranty** on all the requested equipment. The warranty period shall start from the date of final acceptance of the goods, following installation and commissioning, and completion of the training at RDCBSM.

The warranty shall cover:

- (a) Defects in materials, workmanship, and software functionality.
- (b) Attending to intimations of faults in functioning, at its own expense, within seven (7) days of receipt of notice to meet and agree with representatives of RDCBSM the action required to correct the deficiencies.
- (c) Free calibration and validation services at annual intervals during the warranty period.
- (d) Software updates, firmware patches, and compatibility upgrades necessary for continued compliance with EN:15194.

For replacements under warranty, all related costs (transport and other) must be paid by the contractor.

Bidders should indicate the procedures when invoking this warranty. Warranty should be transferable from UNIDO to the end-user/project beneficiary and will be effective on the date of final acceptance of goods/services as per Certificate of Acceptance signed by all parties (UNIDO supplier and end-beneficiary).

G- Quality Assurance

Contractor warrants that all equipment supplied shall be merchantable and fit for purposes for which they are intended and shall conform to the Technical Specifications stipulated herein and shall in all respects conform to and comply with all applicable rules, regulations, standards and shall be of good quality, free from any defects in materials, workmanship, design and performance.

H - Documentation

The contractor is expected to provide the following documents:

- 1) Inspection and trial run report at manufacturer's site/plant prior to shipping.
- 2) User Manual in English (Soft & Hard Copy) x 2.
- 3) Maintenance Manual in English (Soft & Hard Copy) x 2.
- 4) Electrical Circuit Diagram x 2.
- 5) Copy of Software should be provided in a Storage Device as Back Up x 1
- 6) Standard Spare Part List with Part No. list x 1 set.
- 7) An overview of pre-installation requirements (if required):
- 8) Detailed drawings of the works to be completed (as built drawings) showing the full and final details of civil and utility work required for installing the equipment. These drawings shall be suitably indexed and shall be in standard sizes. These drawings shall also be submitted in soft form.
- 9) A brief report of the installation, commissioning and training at RDCBSM.
- 10) Calibration certificate for the equipment after installation at the site.

All requested documents should be provided to UNIDO in electronic copy, consisting of the following electronic files/formats.

- PDF file
- Original work files (WORD, POWER POINT, EXCEL, etc.).

The electronic copy can be provided via e-mail, if the nature and size of the files make it possible. Otherwise, the electronic copy should be provided through cloud storage application as per UNIDO's suggestion.

All documents should be provided in English language.

I - Spare parts and Software support

The availability of spare parts and support for Software issues for equipment to be included in the scope of supply of this TS should be assured for at least ten (10) years (or as per industry standard) post warranty and deviating duration should be indicated by the bidder. In case that the spares are not available in India, the nearest location where they are available must be indicated. Spare parts must be delivered within 12 days from the order. Software support could be made available remotely in case of any trouble shooting and correction.

PLEASE NOTE THAT BIDDERS ARE REQUIRED TO SUBMIT OFFERS IN THE PROVIDED FORMAT OF THE SCOPE OF SUPPLY TABLE IN ANNEX I

The table provided in the Annex I details the specification of supply. The bidders shall complete it by filling in unit and total prices as well as indicate their compliance of deviation from the technical requirements of the Terms of Reference.

The specifications outlined in this document are the minimum requirements to be accepted. Products offered in response to this tender must meet or exceed all requirements herein.

4. Deliverables and General Time Schedule

All activities/deliverables shall be completed and all stated payment supporting documents shall be submitted to UNIDO no later than six (06) months from the date when the contract is signed by the contractor.

Following **activities/deliverables** are expected from the contractor:

No.	Item(s)	Timeline
1	Delivery of Equipment, fixtures and standard spare parts as per scope of supply specified in Annex 1	Within 04 months of signing of contract
2	Installation, commissioning, operationalization of equipment supplied under this ToR along with training (as per details specified in section 3)	Within 06 months of signing of contract
3	All requested documentation as per section 3	Within 06 months of signing of contract

5. Contractor's Personnel

The supplier must appoint/nominate a Project Manager. This Project Manager will be the primary contact point for UNIDO and will be responsible for the services of the supplier and especially for the installation. All staff deployed must be suitably qualified and in possession of the necessary valid permits/visas to work in India for the duration of the contract. In addition, a technical representative of the supplier shall be deployed for the period of installation, commissioning, operationalization of and training for the equipment at the site, as per mutually agreed terms.

A technical representative of the supplier shall be deployed for the period of operation of the installation, commissioning and operationalization of the equipment at the site – RDCBSM, Ludhiana, as per mutually agreed terms. All associated travel-related costs of such representative(s) shall be borne by the supplier.

The bidder is expected to provide CVs of the assigned personnel to be made available for this contract. The bidder is also to indicate where the assigned personnel will be based.

The Project Manager assigned by the supplier as the main responsible for communication with UNIDO shall be available during all working days of UNIDO. For emergencies arising while the Supplier's offices are closed, the Supplier shall make all necessary efforts to be available through emails and phone.

6. Accident or injury to personnel

During installation and commissioning of the equipment at RDCBSM site, UNIDO shall not be liable for any damages or compensation payable by law in respect or in consequences of any accident or injury to any personnel or other person in the employment of the contractor or any of his sub-contractor. The contractor shall indemnify and keep indemnified UNIDO and RDCBSM against all such damages and compensation and against all claims, demands, proceedings, costs, charges and expenses whatsoever in respect thereof or in relation thereto.

7. Compliance with rules and regulations

The contractor shall confirm in all respects with the provisions of any such Statute Ordinance of Law and the Regulations of Byelaws of any local or other duly constituted authority which may be applicable to the works and with the rules and regulations of public bodies and companies and shall keep UNIDO indemnified against all penalties and liability of every kind for breach of any Statute Ordinance of Law, Regulation or Byelaws.

The Contractor shall be responsible for obtaining and maintaining all necessary permits, certifications, and regulatory approvals required for the import, installation, and operation of the equipment into India.

8. Patents, Contractor's liability & compliance with regulations:

The contractor shall protect and fully indemnify UNIDO from any claims for infringement of patents, copyright, trademark or the like. Contractor shall also protect and fully indemnify UNIDO from any claims from Contractor's workforce/employees; their heirs, dependents, representatives etc. or from any other person/persons or Bodies/companies etc. for any act of commission or omission while executing the order. Contractor shall be responsible for compliance with all requirements under the laws and shall protect and indemnify completely UNIDO from any claims/penalties arising out of any infringements and indemnify completely UNIDO from any claims/penalties arising out of any infringements.

9. Language Requirements

All communication with UNIDO will be in English. The provision of drawings, catalogues, illustrations, printed specifications, operating manual and other documentation related to the present project shall be in English.

10. Location

The delivery and associated activities (i.e. installation, commissioning, training and operation) will be carried out at RDCBSM, in Ludhiana, Punjab, India.

11. Payment Terms

The timeline and payment terms based on the deliverables, as well as documents requested for each payment to be processed, are presented in the following table:

No.	Deliverables	Documents required for payment to be processed	Expected completion	Payment
1.	Upon signature of purchase order	a) Countersigned purchase order b) Invoice c) Completed Bank Information Form (BIF)	Upon signature of contract	10% of equipment value
2.	Shipping of all items under order	a) Invoice b) Packing list c) Multimodal bill of lading d) Inspection and trial test reports e) Insurance certificate/policy	Within 04 months after signing of contract	70% of equipment value + shipping cost
3.	Final acceptance following Delivery, Installation, and Commissioning of equipment with the associated fixtures and training as well as all requested documentation as per scope of services in section 3 of the ToR	a) Invoice b) Certificate of acceptance c) Signed training sheets.	Within 06 months after signing of contract	20% of equipment value + remaining costs related to installation and commissioning

Payments will be payable within 30 days upon receipt and acceptance of deliverable and invoice (electronic version) indicating the contract number and instalment requested.

12. Qualification Requirements

Following are qualification requirements for bidder's technical offer to be considered:

a) Corporate registration

The bidder should provide a copy of their **Certificate of Incorporation** or other documents setting forth the legal basis of the company and therefore proving a legal capacity to enter into a contract.

b) Certification of operations to provide requested equipment and services

The bidder should provide a **copy of ISO 9001:2015** or its equivalent attesting quality system certification.

c) Minimum number of years in business

The bidder should be able to give evidence of the **existence of operations of at least five (05) years**.

Individual companies forming Joint Ventures and/or consortia would be also considered.

d) Commitment to ethical principles

Duly filled and signed Statement of Confirmation provided herein.

Failure by the bidder to meet mandatory qualification requirements will exclude the bidder from further consideration in the procurement process.

13. Evaluation Criteria

The bidder should meet following evaluation criteria:

13.1 Technical evaluation criteria

a) Technical offer

The bidder should submit a technical offer in compliance with the Technical Specifications, as well as detailed description/technical sheets/catalogues of equipment, fixtures and spares being supplied, and sub-activities envisaged (installation, commissioning and training).

b) Existence of a local partner in India

The bidder should have a service partner in India and/or own office branch which will serve the after-supply services to RDCBSM, Ludhiana with spare parts and consumables besides services for repair and troubleshooting during break-down of the equipment.

c) **At least three (3) references** to demonstrate status of recognized manufacturer or authorized supplier/dealer of industrial/engineering equipment) with specific experience in safety/electrical testing equipment, along with experience of supply to engineering industries.

d) Delivery timeframe

The bidder should submit a proposed Gantt chart showing start and finish days for various activities forming part of the execution of the contract/order and identifying the delivery dates to this activity schedule. Time of delivery as mentioned in the Purchase Contract/Order shall be essence of the contract and no variation shall be permitted.

e) Language requirements

All communication with UNIDO will be in English. The provision of drawings, catalogues, illustrations, printed specifications, operating manual and other documentation related to the present project shall be in in English.

f) Warranty Terms

The bidder should confirm adherence to warranty requirements mentioned in Section 3-VI). Replacement or rectification, if any, as required at the site shall be notified of the contractor giving full details of the deficiencies. The contractor shall, at its own expense, meet and agree with representatives at site for the required action to be taken to correct the deficiencies. The supplier shall take the requisite action within seven (07) days of receipt of such notice.

Evaluation of technical criteria, and therefore a decision whether an offer is considered technically compliant, will be done according to the compliance or non-compliance against the stated requirements in this technical specification.

Bidders should note that only technically compliant offers/proposals should be further considered for commercial evaluation.

13.2 Commercial evaluation criteria

a) Price and Financial offer

The bidder should submit a financial offer in **United States Dollar (USD)** in format shown under Annex II - Financial Proposal Format. Bidders must have an account in the currency in which they provide their financial offer, or they should provide a note that financial offer in USD is for reference purposes only and the currency in which payment will be expected.

In specific cases where payment is to be made in a currency other than USD, the United Nations Rate of exchange prevailing at the time of payment shall be applicable. For example, bidders from India may indicate their proposal in INR, along with breakdown of applicable taxes.

The financial offer should contain all costs involved in providing the required equipment, fixtures, spares and associated services specified in these Terms of Reference, including but not limited to:

- Itemized cost of each item/equipment including standard spares and fixtures.
- Delivery costs.
- Insurance costs.
- Training costs.
- Installation and commissioning costs.

b) Acceptance of UNIDO terms and conditions

- the bidder shall confirm the acceptance of UNIDO Model Contract and UNIDO General Terms and Conditions of Contract;
- the bidder shall confirm acceptance of proposed UNIDO Payment Terms. including payment terms.

The contract shall be awarded to the qualified bidder whose Technical Offer has been found substantively responsive and whose Commercial Proposal is the lowest cost to UNIDO.

14. Submission of Offers

Bidder's offer must be signed by an official legally authorized to enter contracts on behalf of the bidder.

Together with an offer, bidders must submit a statement that bidder's **quotation is valid for a minimum period of 120 days** from the date of the quotation. Once the offer is accepted during this period, the price quoted must remain unchanged for the entire period of the resulting contract.

15. Award conditions

UNIDO reserves the right to split an award between any suppliers in any combination, as it may deem appropriate. If the quotation is submitted on an "all or none" basis, it should be clearly stated as such in your response to this RFQ.

ANNEX I - SCOPE OF SUPPLY

Testing equipment with required accessories/fixtures for meeting electrical safety standards and performance requirements of the e-bicycles and its sub-assemblies as per EN 15194 – Scope of Supply includes Delivery, Installation and Commissioning and Training

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
1	<u>Battery Analyser</u> Purpose: perform Battery pack analysis as per EN15194 for safety & endurance a) Charge/discharge cycling to simulate real-world usage and aging. b) Capacity retention and efficiency tracking over time. c) Voltage and current precision for accurate performance profiling. d) Thermal integration with climatic chambers to assess temperature-dependent behaviour. e) Custom load profiles to mimic EPAC usage patterns (e.g., acceleration, hill climbing). Technical Specifications: i. Voltage range: 0V to 100V or $\pm 100V$ ii. Current range: 0.1A to 40A iii. Resolution: $<1\mu V$ (Voltage)-24bits ADC, $<1mA$ (Current)-18bits DAC iv. Control accuracy: $\pm 0.02\%$ for both voltage and current v. Minimum step time: 5ms vi. data logging rate: up to 2000 points/sec per system vii. Circuit type: Bipolar linear circuitry (seamless transition between charge/discharge without delay) viii. Software: Real time plotting, unified desktop/web/API interface for control and data streaming ix. Auxiliary inputs: Thermocouple, thermistor x. Allow thermal integration with third party climatic chambers with DAQ system to test as per EN50604-1/clause 5.2 -Annexure C xi. Safety features: Watchdog circuit, over-voltage/current clamps safety interlocks – built in protection against overvoltage/overcurrent/thermal runaway. xii. Connectivity: TCP/IP (ethernet) xiii. Channels: 4 no. allowing modular scalability xiv. Input voltage: 240V xv. Each channel operates independently allowing precise charge/discharge cycles xvi. Cell and battery pack holders / fixtures of an EPAC - 36V / 48V packs xvii. Include any other necessary standard spares and fixtures and specify separately.	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
	xviii. General Appearance a) Rack-mounted or benchtop unit, often housed in a metal enclosure with ventilation grills. b) Front Panel: Four independent channel interfaces, each with its own set of terminals (positive/negative) for connecting battery cells or modules. ➤ LED indicators or small displays for channel status. ➤ Safety fuses or circuit breakers per channel. c) Rear Panel: Power input, communication ports (Ethernet, USB, CAN), and possibly cooling fans. Optional integration ports for environmental chambers or auxiliary sensors. d) Cooling: Please specify for example - forced air or liquid-cooled depending on current rating. xix. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set					

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
2	<u>Climatic Chamber</u> Purpose: To validate durability and safety of EPAC a) Environmental stress testing – temp., humidity and thermal cycling b) Battery and electronics validation-battery packs, controllers and displays, wiring & connectors c) Humidity bias testing evaluates degradation of connectors and PCB coatings under prolonged exposure" Technical Specifications: i. Temperature range: -20 deg. C to +150 deg. C ii. Humidity range: 10% to 98% RH iii. Ramp rate: up to 5 K/min or 5 deg. C/min. (Environmental Stress Screening variants) iv. Volume of chamber: 1000L (1 m3) v. Control Accuracy: $\pm 0.1K/C(temp)$ and $\pm 2\%$ RH (humidity) vi. Refrigerant: Eco friendly like R744(CO2) or R469A etc. vii. Access ports: diam. - 80 mm or larger (min. 2 ports) stainless steel feedthroughs (sensor routing, battery leads, CAN Bus). viii. Safety features: fire-rated housing, Triple test material fuse, Pt sensors, fault triggered shutdown, door interlock. ix. Software interface: offer system for remote control, logging and CE documentation for example like WEBseason+SIMPATI or any other proprietary software etc. x. Load capacity: up to 250 Kg with 50Kg per grid shelf. xi. Allows for integration with third party battery analyser for thermal abuse protocols. xii. Include standard spares (gaskets, door seals, filters, fuses, relays, sensor probes). xiii. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
3	<u>Insulation resistance tester (Megohmmeter)</u> <u>Purpose:</u> Verify electrical safety and insulation integrity Technical Specifications: i. Test Voltages: 250V, 500V and 1000 V DC ii. Voltage measurement: AC up to 750V and DC up to 1000V iii. Resistance range: 0.1 MΩ to 2 GΩ iv. Accuracy: +/- 4% v. Safety features: High voltage buzzer and anti-reverse protection vi. Display: 3.5-inch backlit LCD with data hold vii. Certification: CE marked viii. Dimensions: 17x 111 x 6 cm ix. Weight: ~ 200 gm x. Calibration certificate (2 years) and user manual (English) xi. Warranty certificate (2 years) xii. Include standard spares (test leads, fuses, external voltage adapter, software suite). xiii. Power source: Battery powered	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
4	Spray and splash test chambers Test Standards Supported: EN 15194, IS/IEC 60529 IPX3 and IPX4/IPX5/6/6K Purpose: Water ingress testing for IPX4/IPX5/6/6K compliance Technical Specifications: i. Swivel tube standard Radii: 200 mm ii. Swivel tube drive Type: Pneumatic or motorized swivel tube iii. Swivel tube spray angle: $\pm 60^\circ$ to $\pm 180^\circ$ adjustable iv. Water flow rate: ~ 10 L/min (IPX4) v. Programmable Timer: 3 seconds to 30 hours vi. Turntable speed: ~ 1 rpm (to ensure uniform exposure) vii. Control Interface: remote access via tablet or smartphone to monitor, control, and program chamber remotely for example like WEBSeason® or any other proprietary software. viii. Cycle Customization: Fully automated with adjustable flow rate and spray duration. ix. Test Space volume: 1015 litre (desirable) x. Test space dimensions: (W x H x D) 1000x1015 x 1000(mm) (desirable) xi. Drip Area for test IPX1/2 (W x D): 400 x 400 (mm) xii. Swivel tube radius for IPX3/4 tests: 200 mm xiii. Radiation tunnel for IPX5/6/6K tests (WxHxD): 670 x 180 x 2900 (mm) xiv. High-pressure equipment for IPX9/9K tests (WxHxD): 1500 x 1900 x 1900(mm) xv. Stainless steel test chamber xvi. Digital flow measurement xvii. Operation with web panel - xviii. Test chamber lighting with a LED lamp xix. 100 mm access port in the side panel xx. Water recirculation system to reduce consumption and to support sustainability. Drainage & collection tray (stainless steel base with water outlet for safe disposal). xxi. Specimen mounting fixtures (adjustable racks/brackets for securing EPAC components- battery, motor, connectors). xxii. Include standard spares (nozzles, bearings/seals, gaskets/seals, filters, fuses, relays)	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
	xxiii. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set					

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
5	<u>Drop tester</u> Purpose: To check safety, durability, and regulatory compliance of batteries used in EPAC as per EN 15194 a) Assess mechanical integrity and safety under impact. b) Identify risks like leakage, thermal runaway, or explosion Technical Specifications: i. Drop height: 300 to 1500 mm ii. Drop Orientations: flat, edge and corner iii. Drop floor material: A3 steel plate - 500 x 600 x 200 mm(approx.) iv. Control system: pneumatic or motorized with digital display v. Height Accuracy: ± 1 mm or $\leq 2\%$ error margin vi. Max load capacity: 10 Kg (battery packs) vii. Safety features: Hydraulic buffer, rail guards and auto rest viii. Approx equipment dimensions - 700(W) x 900 (D) x 1650 mm (H) ix. Approx. Control box size: 210 x 180 x 140 mm x. Power supply: AC 220 V, single phase, 5 A xi. Include standard spares(fuses, relays, shock absorber pads, replacement fixtures/clamps) xii. Other features: a) Single arm, double column structure b) Electric reset, drop and lift mechanisms c) Hydraulic buffer system. xiii. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1 set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
6	2W Chassis Dynamometer Purpose: To measure force, torque, speed, and power output of an EPAC as per EN15194 a) Simulates real-world riding Conditions b) verification of the power assist ratio c) Battery range & efficiency Testing-validates claimed Watt-hour capacity by simulating a full ride cycle under load. d) Speed limitation checks at 25km/hr cut off. e) Durability & endurance Trials Technical Specifications: i. roller speed: up to 150 km/hr, speed to step-up in least count of ± 0.5km/hr ii. roller dimensions: approx. 530 x 300 mm width iii. surface material: steel with hard chrome plating iv. Base Inertia: 90 Kg simulates EPAC and rider weight v. Simulated inertia range: 80 - 260 Kg, Accuracy: $\pm 1\%$ vi. Torque Meter : TQ-1206 flange type Torque curve mapping is performed by applying incremental pedal force and measuring motor output across RPM ranges. vii. Drive simulation: pedal shaft drive unit(manual/auto) viii. AC type Dynamometer power: 7.5 Kw rated ix. Speed range: 150 - 1500 rpm x. Detection Method: Strain gauge xi. Telemetry: Non-contact signal transfer xii. Control system: Flexible automatic measuring system (FAMS)-F5 or equivalent (modular software suite), assist ratio, cut off, range per charge xiii. Driver's aid display: Graphical speed and gear shift guidance for ensuring repeatable test cycles xiv. Cooling system: axial flow fan xv. Accuracy: $\pm 1\%$ across torque, speed, and inertia xvi. Automatic wheel clamping using pneumatic system xvii. Include standard spares (replacement torque meter seals, calibration weights, encoder cables and connectors, fuses, relays, control board components, roller surface cleaner kit). - xviii. Manuals, Software and Spare Part list:	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
	a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set					

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
7	<u>Electrodynamic shaker</u> Purpose: Validate mechanical durability under sinusoidal vibration Test Battery, ECU, Display, Motor, Frame of EPAC Technical Specifications: i. Sine force (peak): 1000 Kgf (10 kN) ii. Random force (RMS): 10 kN iii. Shock force (peak): 20 kN iv. Acceleration (Sine): 75-90 g v. Acceleration(random): 70g vi. Acceleration (shock): 160 g" vii. Velocity (peak): 1.7 m/s to 2.5 m/s viii. displacement (peak to peak): 38 mm to 51 mm (1.5 in to 2 in) ix. Useful Frequency range: 5 to 3200 Hz x. Axial resonance: ~ 2800 Hz ±5% xi. Armature diam.: 240 mm xii. moving element mass: 12 Kg - 17 kg xiii. max. dynamic payload capacity: 240 - 260 kg (Max weight of test specimen the shaker can vibrate effectively) xiv. Shaker body mass: 1500 Kg xv. Stray magnetic field: < 1.5 mT (15 gauss) xvi. Suspension axial stiffness: < 1 Kg/mm xvii. Suspension cross axial stiffness ; 300 Kg/mm - xviii. cooling system: air cooling to meet varied loads xix. compressed air requirement: 6 bar (87 psi) xx. Amplifier model: DSA 10K xxi. Power output: 10 KVA xxii. Input sensitivity: 4 Vrms (compatible with standard controllers) - xxiii. Input impedance: > 10 KΩ xxiv. Signal to noise ratio: 80 dB xxv. Total harmonic distortion(THD): ~0.5% xxvi. Power efficiency: 92% - xxvii. Include standard spares if any - xxviii. CE certified	1				

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
	xxix. protection interlocks: over/under volage, over current, phase loss, over temp., cooling failure and user-friendly interlocks. xxx. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set					

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
8	Dust Chamber Compliance Standards : EN 60529 (IP5X/IP6X), IEC 68-2-68 Purpose: Applicable for EPAC components testing for IP5X/6X standard Technical Specifications: i. Dust pre-heating mechanism for free flow of the dust ii. Oil free vacuum pump to simulate negative pressure for IP6X test iii. Volume: 1000 L iv. Approx dimensions: 1000-Width x1000-Depth x1000- Length in mm(inner) and 1200x1200x2300 mm (outer) v. internal material: Internal test chamber is manufactured with stainless steel vi. external material: min. 2mm thick CRCS with powder coated panels/door vii. Thermal insulation: Internal test chamber and outer body is isolated using mineral wool or comparable material viii. Dust hopper and circulation Blower: Free flow dust circulation centrifugal blower ix. working temp. 25-30 deg. C and 10-90 % RH x. Controller: Programmable timer to set the test duration xi. Standard Accessories & fixtures: a) Mounting fixtures for battery housings, connectors and motor enclosures (EPAC components) b) Spare gaskets and filters c) Standardized test dust (ISO 12103-1 A2 Fine Dust) – 1 set xii. Manuals, Software and Spare Part list: a) User Manual in English (Soft & Hard Copy) x 1 set b) Maintenance Manual in English (Soft & Hard Copy) x 1set c) Electrical Circuit Diagram x 1 set d) Copy of Software should be provided in a storage Device as Back Up x 1 set e) Calibration certificate (2 years) f) Warranty certificate (2 years) g) Spare part List with Part No. x 1 set	1				
I	Sub-Total, EX-Works					

			TO BE COMPLETED BY THE BIDDER			
Item	Name and Required Parameter of the Equipment	QTY	Unit price USD	Total price USD	Compliance yes/no*	Remarks
II	Cost of transportation (including insurance) to the Site DAP Incoterms 2020					
III	Installation, commissioning and operation at site					
IV	Training at site					
V	GST					
	TOTAL PRICE					

* Compliance must be confirmed in detail by the bidder's offer and technical documentation and will be verified by UNIDO during technical evaluation; if not compliant with UNIDO parameters, the bidder must indicate his parameters in column 'Remarks'.

